



Review article

The efficacy of exercise in patients with myalgic encephalomyelitis/chronic fatigue syndrome: A systematic review and meta-analysis

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ABSTRACT

Objective: To evaluate the efficacy of exercise training for patients with myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS).

Methods: Seven databases were searched from inception to December 2025 for randomized controlled trials (RCTs) on exercise effects in ME/CFS. A random-effects model was employed to calculate the pooled effect size, with Q , I^2 statistic, and τ^2 assessing a heterogeneity. Additionally, we performed moderator analyses to explore stability of results.

Results: A total of 17 RCTs ($n = 1944$) were included. Exercise interventions significantly improved fatigue (16 RCTs, $n = 1673$; standardized mean difference [SMD] = 0.85, 95% CI 0.65 to 1.06, $I^2 = 85.3\%$), sleep quality (5 RCTs, $n = 806$; SMD = 0.42, 95% CI 0.31 to 0.53, $I^2 = 41.2\%$), QoL (11 RCTs, $n = 1171$; SMD = 0.99, 95% CI 0.73 to 1.26, $I^2 = 89.7\%$), depression (9 RCTs, $n = 1279$; SMD = 0.56, 95% CI 0.43 to 0.69, $I^2 = 45.8\%$), and anxiety (9 RCTs, $n = 1279$; SMD = 0.14, 95% CI 0.04 to 0.24, $I^2 = 21.2\%$). However, the effects on cardiopulmonary function (3 RCTs, $n = 150$; SMD = 0.13, 95% CI -0.11 to 0.43, $I^2 = 0.0\%$) and functional capacity (4 RCTs, $n = 459$; SMD = 0.09, 95% CI -0.21 to 0.38, $I^2 = 47.4\%$) were not significant. These results may be influenced by potential moderators such as intervention length, frequency, session duration, and medication use. Crucially, there was no significant difference in acceptability or adverse events between two groups.

Conclusion: Exercise significantly improves fatigue, sleep, QoL, and depression in ME/CFS patients. However, exercise does not yield significant improvements in cardiopulmonary or functional capacity. While not a cure, exercise is an effective strategy for symptom management, provided that individual energy limits are strictly respected to prevent harm.

1. Introduction

Myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS) is a debilitating and chronic disease [1]. It is characterized by persistent and profound fatigue, post-exertional malaise (PEM), unrefreshing sleep, and cognitive dysfunction [2]. Furthermore, depression, anxiety, and significant impairments in quality of life (QoL) are frequently observed in patients with ME/CFS [3]. The disease is widespread, with a global prevalence ranging from 0.2% to 2.8%, impacting populations across all ethnic, socioeconomic, and age groups [4]. Moreover, the economic burden of ME/CFS is substantial, with an estimated annual cost of 36

billion to 51 billion dollars in the United States alone [5].

For decades, extensive research across diverse chronic populations—such as individuals with cardiovascular disease, diabetes, and cancer—has firmly established exercise training as a highly effective modality for mitigating fatigue and enhancing QoL [6–9]. Theoretically, these benefits are mediated through multidimensional pathways: biologically, by improving mitochondrial metabolism [10,11]; psychologically, by facilitating autonomic regulation and reducing stress [12,13]; and socially, by fostering interaction [14]. Given that profound fatigue is the defining symptom of ME/CFS, extending these proven exercise interventions to this population is of significant clinical interest. However,

Abbreviations: CIs, Confidence intervals; GRADE, Grading of Recommendations Assessment, Development and Evaluation; ME/CFS, Myalgic encephalomyelitis/chronic fatigue syndrome; NICE, National Institute for Health and Care Excellence; PEM, post-exertional malaise; QoL, Quality of life; RCTs, Randomized controlled trials; RoB, Risk of bias; RR, Risk ratio; SDs, Standard deviations; SMDs, Standardized mean differences.

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